

Tonight is
Carb City's Star Festival!



The Energy Heart
(能量之心) has stopped!



Before the moon rises,
I'll bring the light back!



MISSION: Pass the three
carbohydrate gates (碳水化合物關卡).



1 MONOSACCHARIDES (單糖) = ONE sugar unit



MONO GATE



Glucose (葡萄糖)
— candy, quick energy



Fructose (果糖)
— fruit, sweetener



Galactose (半乳糖)
— part of lactose

MEMORY SEAL
MONO
= ONE



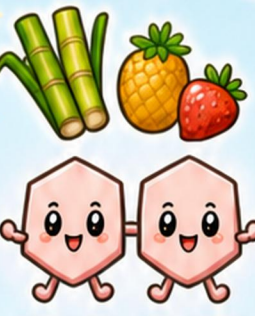
2 DISACCHARIDES (雙糖) = TWO sugar units



DI GATE



Maltose (麥芽糖)
— malt, energy



Sucrose (蔗糖)
— sugarcane and fruit

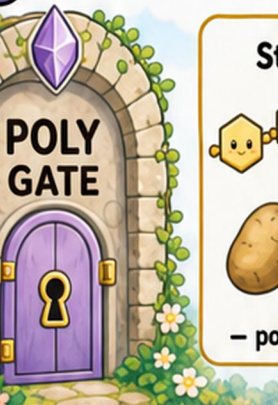


Lactose (乳糖)
— milk sugar

MEMORY SEAL
DI
= TWO



3 POLYSACCHARIDES (多糖) = MANY sugar units



POLY GATE

Starch (澱粉)




— potato and grains

Cellulose (纖維素)




— vegetables

Glycogen (肝糖)




— liver and muscles

MEMORY SEAL
POLY
= MANY

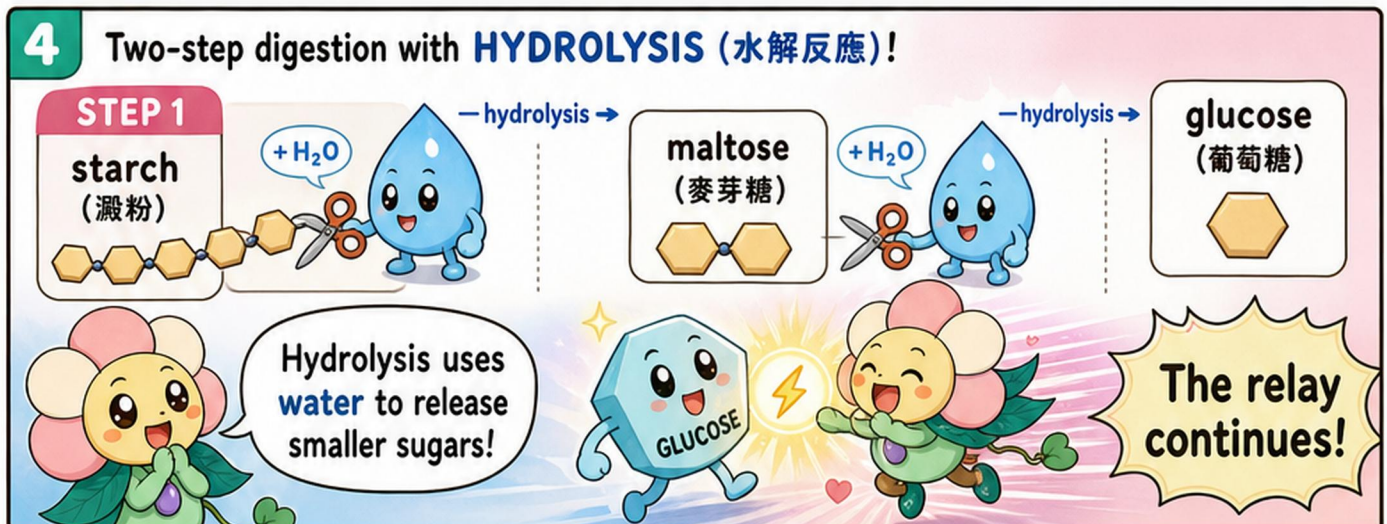
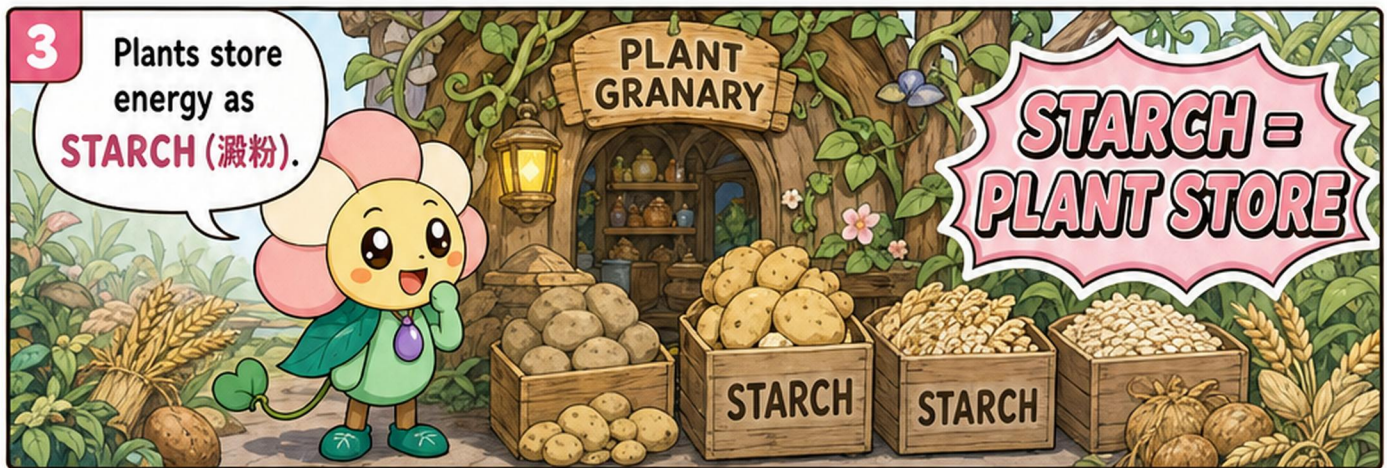


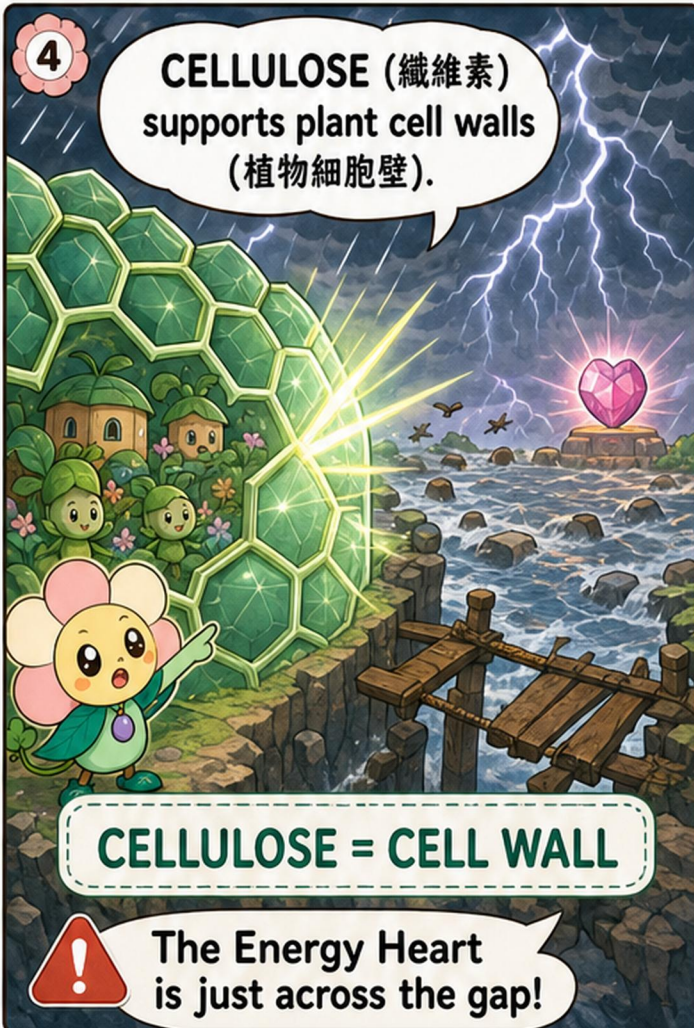
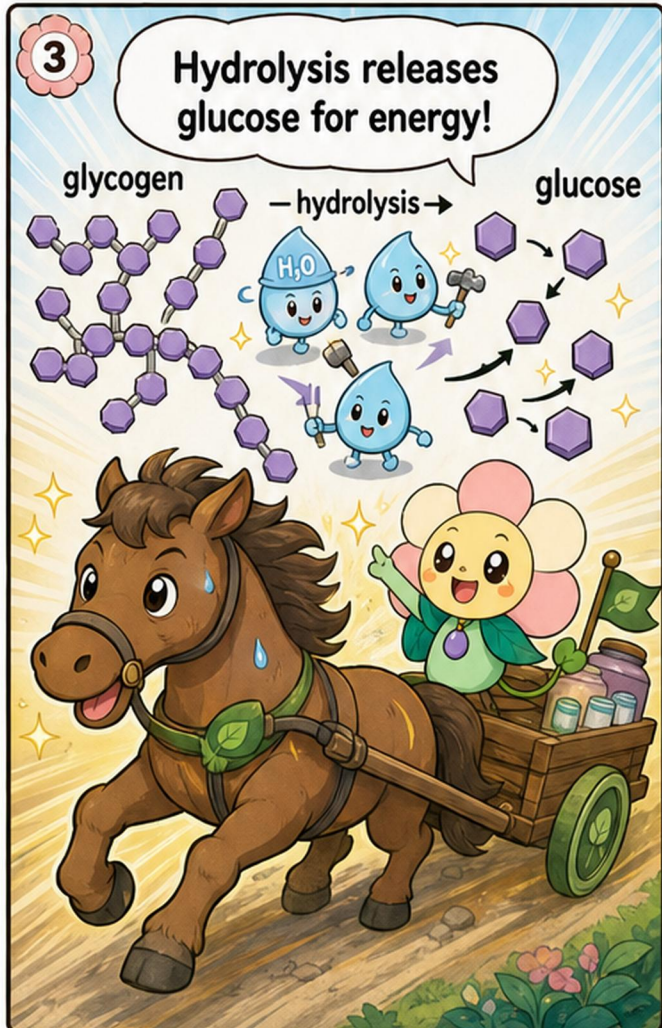
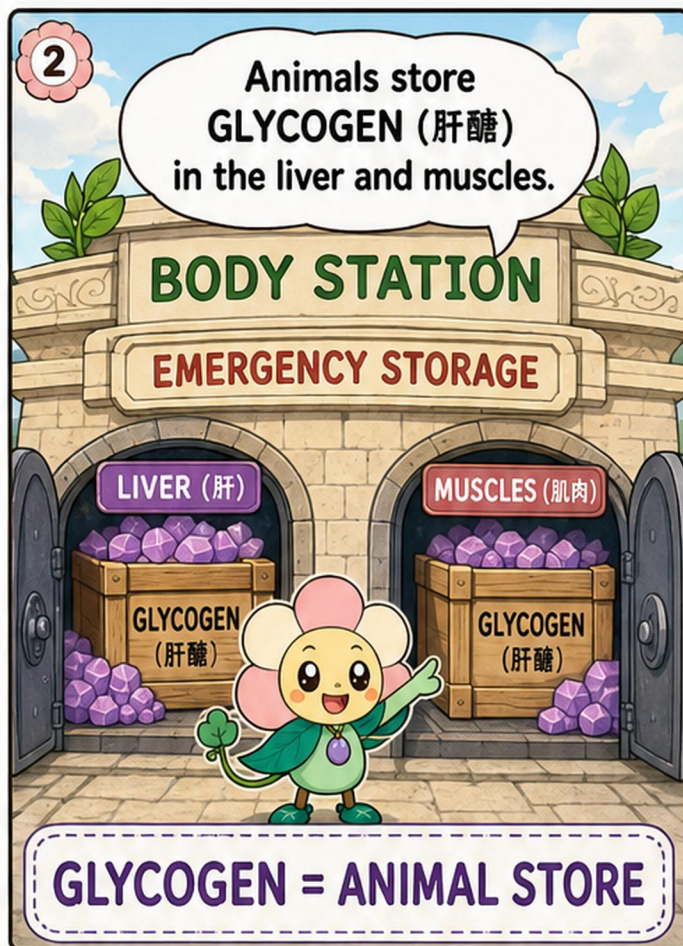
Three separate examples — not joined together.

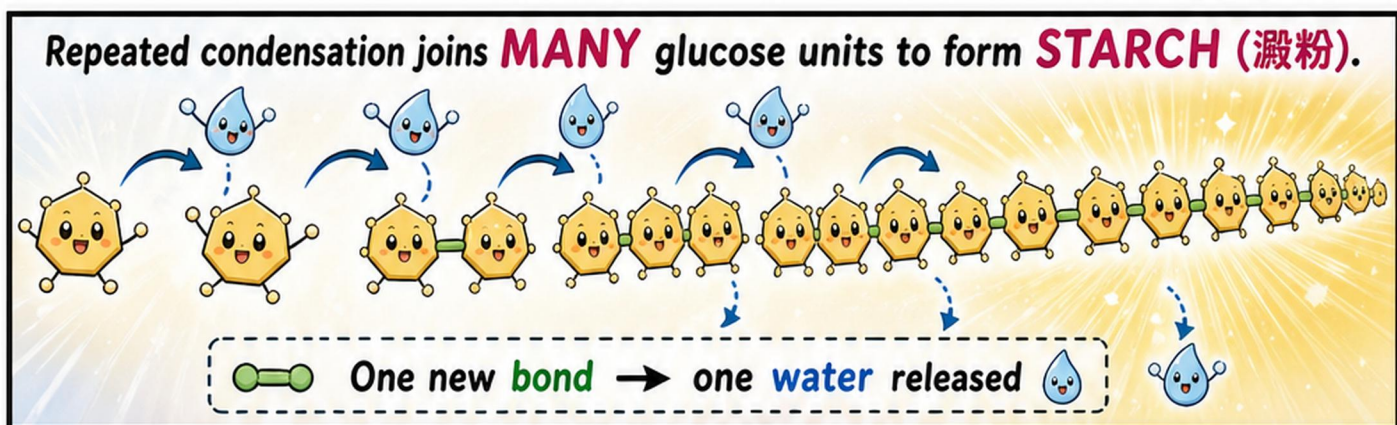
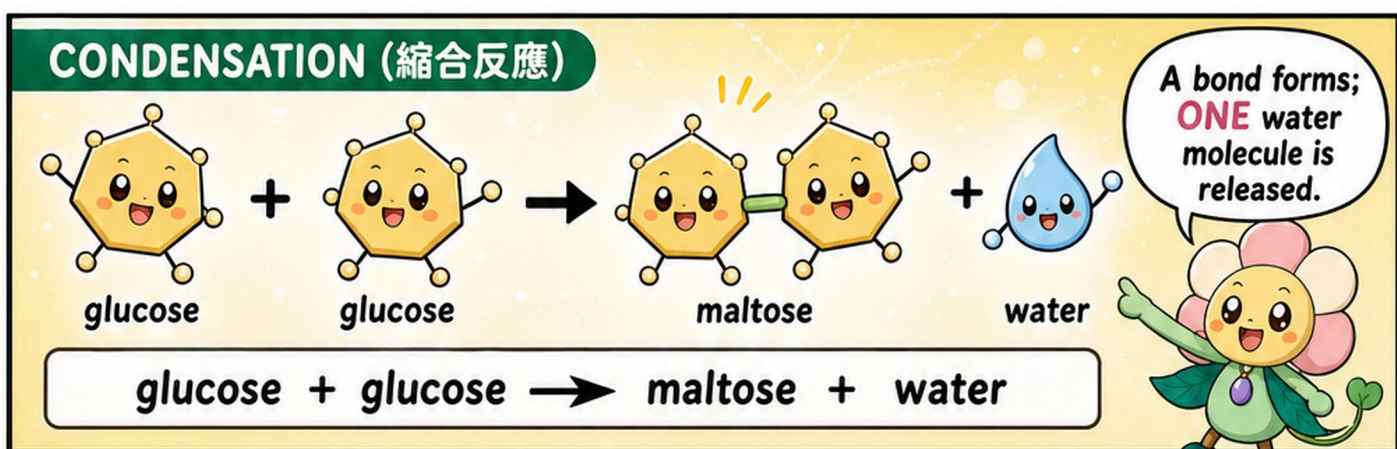
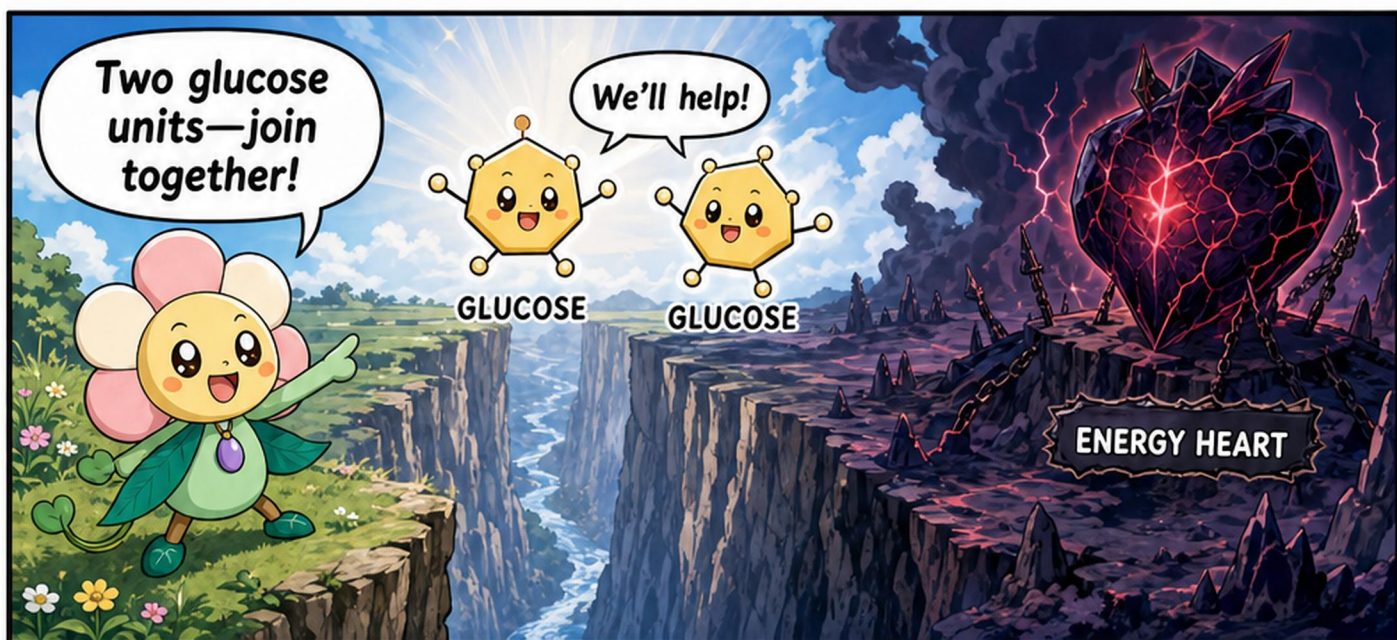


One, two, many—the path is open!

NEXT:
The carbohydrate energy cycle!









HYDROLYSIS (水解反應)

$$\text{GLUCOSE} - \text{GLUCOSE} + \text{H}_2\text{O} \rightarrow \text{GLUCOSE} + \text{GLUCOSE}$$

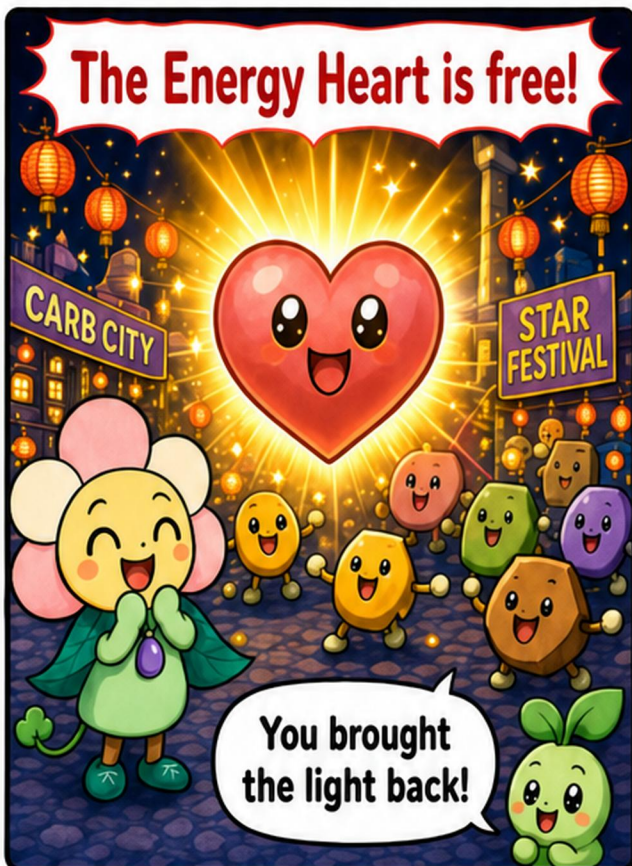
MALTOSE WATER GLUCOSE GLUCOSE

Use **ONE** water molecule to break **ONE** bond!

GLUCOSE

GLUCOSE

BREAK + USE WATER



MONO	DI	POLY	Condensation: BUILD + RELEASE WATER	Hydrolysis: BREAK + USE WATER
one	two	many		maltose water glucose glucose